*Chapter 9*

# Requirements definition and equipment nomenclature

## 9.1 Overview

The provision of a C&I system for a power station is a complex matter which requires careful and comprehensive administration. The task is demanding: the design of the equipment must be correct, systems must be designed on time, equipment has to be carefully specified and purchased, and everything has to be delivered to site, installed and commissioned to a tight programme which interweaves the C&I system with the many other activities that will inevitably be taking place on site at the same time. When complete the system should be fully supported by comprehensive documentation which enables maintenance staff and users to deal with it.

The actual process of designing the C&I systems forms only one part of the many activities that go together in the task of engineering a complete contract. Although some of the other operations may seem mundane and trivial, they are really anything but that. They are as essential to the contract as the technical design work.

## 9.2 Defining the requirements

The outcome of a major engineering project such as the design, installation and commissioning of the C&I systems of a power station will to a large extent depend on specifying the requirements at the outset so that what is provided fits within the budget and is fit for the intended purpose. Equally exact documentation is required at each stage as the system metamorphosises from the original concept to the final, functional form.

This procedure requires a considerable amount of definition, and the following outline lists the documents that might be required over the lifetime of a typical project, listed in the order in which they may be expected to be generated. This does not pretend to be an absolute definition that must be rigorously followed on every installation. It is a practical system that has produced good results when followed on several projects. Other documentation systems offered by a system vendor may be perfectly acceptable, provided that the same degree of definition is achieved at each stage.

### *9.2.1 The functional specification*

A system can be defined in several different ways, but an essential requirement is the definition of what the system should do *in relation to the process* it is monitoring and/or controlling. This requirement is met by the Functional Specification (FS). This is a process-related definition of the functions that the system will be required to perform. It does not provide detailed descriptions of the system hardware and software, such as response times, power supplies, environment etc., except where these are critical to meeting the functional objectives of the installation. Because it defines the requirements, the FS should be one of the documents against which the vendor is invited to submit a commercial bid for a project.

A typical FS will describe the plant as a whole, and then discuss the control loops with the required accuracy, response times and dynamic range of each loop.

### *9.2.2 The technical specification*

Having said *what* the system should do, it is then appropriate to define, in some detail, *how* the functions should be achieved. This purpose is served by a Technical Specification (TS) which describes the system configuration in terms of the electronics and computer technologies to be employed. Because it defines the technology and facilities required, this document should be requested from the vendor when he submits a proposal for executing a project.

The TS should include the following definitions:

- the size, type and number of operator displays to be provided;
- the overall configuration (heirarchical, distributed etc.);
- method of programming (block-structured, Boolean, ladder-logic etc.);
- the physical environment in which the equipment will be expected to operate (defined in terms of temperature, humidity, vibration, shock etc. as well as dust levels to be encountered, hazardous-area requirements etc.);

- the performance required (speed, response time, availability, memory etc.);
- power supplies available (including voltage and frequency excursions);
- safety requirements (backup, redundancy, fault tolerance etc.).

The TS will also lay down the requirements for testing, such as:

- Factory-acceptance: where the system is set up in the supplier's premises, connected to a simulator, or to switches and signal sources for inputs and to meters and lamps (or LEDs) to show outputs, and then put through a series of routines to show that it performs as required.
- Site-acceptance tests: performed after the system has been installed and commissioned on site, when it is subjected to changes in desired-value settings, simulated equipment trips and so on, to prove that it reacts correctly and in good time to such events.
- Reliability run: where the equipment is left in full control of the process, to demonstrate that is is capable of operating correctly for a defined period, with no malfunctions. The document should state the duration of the reliability run, the conditions under which this test will be expected to operate and what should happen if a failure occurs (e.g. start again and repeat the test).

Any commercial requirements relating to guarantees, performance bonds etc., should be defined separately, although these will interrelate with the TS and should therefore be referred to.

Note that the various acceptance test procedures will at this stage be defined only in very general terms. A full definition is provided by test specifications (as described later).

### *9.2.2.1 Making provision for site tests*

It is usual to retain a sum (typically 5% of the overall contract value) which is paid only when the vendor has proved that the system is capable of performing satisfactorily. However, it should be recognised that a control-system supplier can demonstrate that his equipment and systems are capable of functioning as required only if the plant is made available to him for testing the system's performance on the operating plant. It is unreasonable to retain what may in fact be a substantial sum of money without giving the supplier a reasonable opportunity to prove that his equipment is as accurate, fast-responding and reliable as required. Yet, it is quite common for the plant owner to procrastinate over performing such tests. The reason for this is that a power plant represents a major investment, and starting the recovery of that investment must naturally commence as quickly as possible. As soon as the plant has been completed

to the point where it can start earning its keep, strong commercial pressure comes into play, requiring the plant to operate at maximum output. The pressure will be to start earning revenue as quickly as possible and for as long as possible, to maximise earnings. Reducing the output of the plant for the purpose of carrying out tests is therefore unpalatable.

It is in everybody's interests that this dilemma is recognised and a suitable form of words developed to cover the situation, yet this is not often done. One solution is to relate the financial retention to the test programme, defining when the tests will be performed and adding words to the effect that if the tests are not carried out within a defined time after commissioning has been completed, performance retentions will be released provided that the delay is for commercial reasons outside the control of the C&I supplier.

## 9.3 The KKS equipment identification system

Each item of equipment on a power-plant site has to be identified by a method which enables it to be uniquely defined, specified, purchased, installed, commissioned and maintained, and this requires some logical way of numbering equipment.

Although several systems of nomenclature can be identified, two methods are in widespread use: one American the other European. The latter is knows as the KKS (Kraftwerk Kennzeichen system translated as power station designation system), which was developed by a consortium of large manufacturers and users under the auspices of the German VGB Technical Committee on technical classification systems [1].

KKS nomenclature is extremely comprehensive, and once it has been understood the system provides a very useful method of identifying any piece of equipment and its operational role in a plant. The system has its weaknesses, but it is so widely used, both in Europe and wherever European influences are felt, that it is very important for the control system designer to have at least a rudimentary understanding of its operation.

KKS defines everything on a plant, from the smallest electronic component to the largest turbo-alternator and even covers the buildings that contain it all.

Unfortunately much of the English-language explanatory documentation that is available has been translated from the original German in such a way that the meaning is sometimes obscure, and fine levels of classification have often become blurred. (For example, the distinction between 'ancilliary systems' and 'auxiliary systems' may not be immediately

obvious.) The system also uses alphabetic characters which do not readily relate to English-language names of the equipment or function. For example, valves are prefixed AA, and the designation letter C is used to identify the purpose of an instrument. This can add to the confusion, since many engineers relate C to a controller. Further confusion can be caused because KKS uses the letters DP to indicate a pressure instrument used in conjunction with a control system or DCS, when DP is usually taken as referring to differential pressure.

Those who are used to the American system of nomenclature may find it confusing that the KKS codes for all equipment are determined by the functional area of the plant *to which the equipment relates*. For example, in a steam-temperature loop where the measured variable is the temperature of the steam at the final superheater outlet, the transmitter will be have a KKS code beginning with LBA, but the spray-water control valve, as part of the feed-water system, will be given a code beginning with LAA. This is very different from the American system, which allocates a conzsistent area code *to the whole loop*, and this can be confusing. It is particularly important that this distinction should be clearly understood by plant personnel, since with the KKS nomenclature the tag number allocated to the valve merely indicates that it is handling water, and gives no indication that the valve is controlling the steam temperature. Critics of the KKS system use this as evidence against the system, stating that when the completed installation is in place, the nature of the medium being handled by the valve is fairly obvious (because of the size, nature or colour-coding of the pipework), whereas the effect of the valve is not. With the older system, the fact that it is a spray-water valve is apparent from the tag number.

The differences are illustrated in Figure 9.1, which shows how the two systems are applied to the same type of control loop. It will be appreciated that, on the actual plant, the tag number allocated to this valve will merely identify it as being part of the feed-water system, and it will not be apparent that its controlling effect is on the steam temperature.

### *9.3.1 The importance of agreement and co-ordination*

The KKS system allows different users to adopt different approaches, stressing that all parties involved on the project should agree on the selected path to be followed. Although this does show a degree of flexibility, on a given project it becomes confusing to those who were not party to the decisions that were originally made. If only for this reason, it is important that an agreement on the numbering methodology is reached as soon as work commences on a project. It is equally important that a single individ-

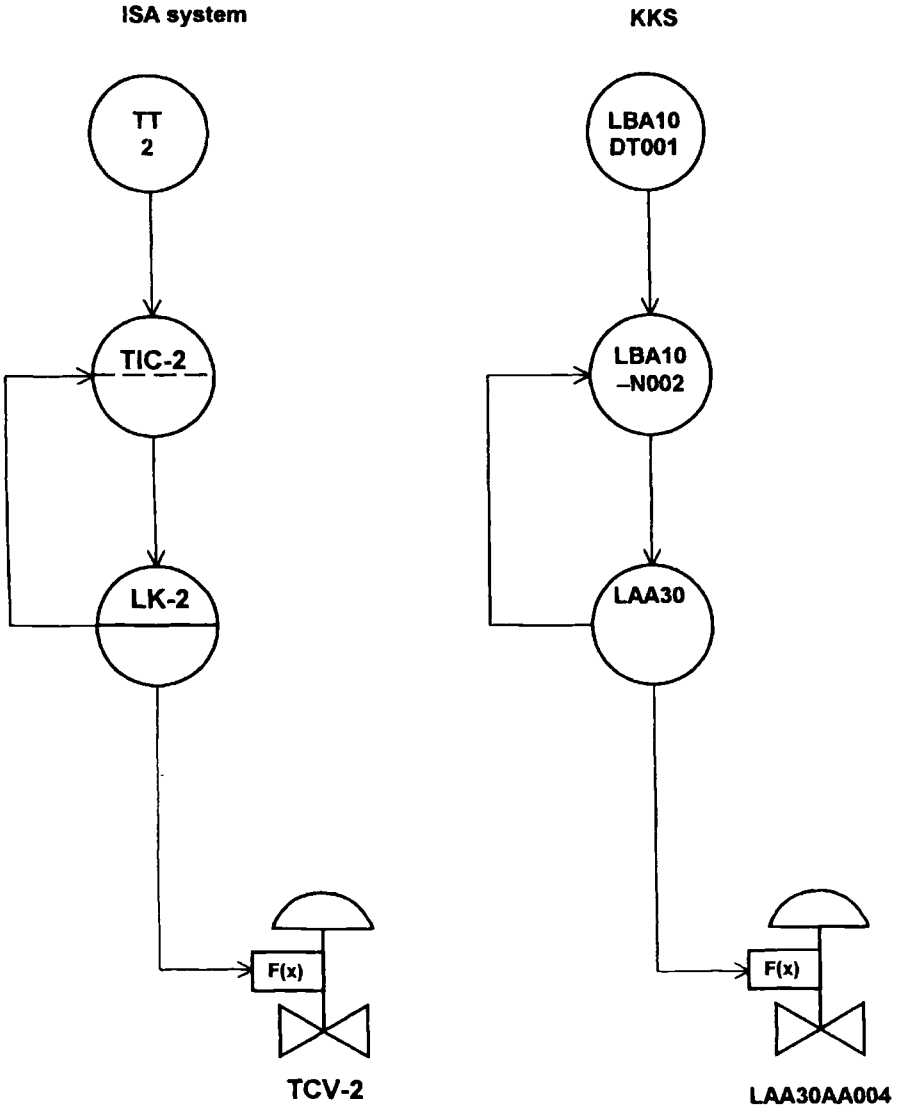



*Figure 9.1 Comparison of KKS and ISA numbering systems applied to a steam-temperature control loop*

ual is appointed to control and co-ordinate the allocation of numbers. This will avoid duplication and different symbols being applied to instruments that serve the same sort of function.

It is also important to understand that the KKS code for an item is almost invariably dictated by the functional area of the plant on which it is

used (an exception is the DCS itself, since this generally serves almost all areas of the plant). Therefore, before a number can be applied to, say, a temperature transmitter, it is necessary to know the KKS code for the pipe to which it is fitted. This dependence renews the pressure for the plant piping and instrumentation diagrams (P&IDs) to be defined at an early stage in a project and for them to be as accurate and complete as possible before work begins on specifying the instrumentation and systems. This point cannot be overemphasised. Problems will inevitably arise as soon as any attempt is made to allocate KKS numbers to instrumentation before the P&ID stage has been completed.

It must be recognised that the KKS system offers a comprehensive method of housekeeping which can greatly enhance the task of allocating nomenclature to all equipment on a plant. This is a routine, but surprisingly costly, facet of the administration of a project. Properly used, KKS can yield considerable economies and efficiency: poorly administrated it becomes a nightmare.

### *9.3.2 Review of KKS*

The following is an attempt to explain how KKS identifies the components that go to make up the C&I systems (and *only* the C&I systems) of a power plant. The structure of this book does not allow for a comprehensive guide to be provided, but the following brief outline should at least enable the reader to grasp the principles.

#### *9.3.2.1 The levels of coding*

The number allocated by the KKS system to a piece of equipment is broken down into a number of sections or 'levels' (see Figures 9.2 and 9.3) and within each of these levels is a field or set of fields, each field being occupied by a letter or number. Each letter or number is allocated a field identification: G for the plant, F for the function, A for the equipment and B for the component.

- The Level 0 code is used where there are, for example, two power plants on the same site. These are usually designated as 'A' and 'B', but the system allows numeric characters to be used if desired. If only one plant exists on the site the first character is omitted altogether.
- The first digit of the Level 1 code identifies the boiler/turbine unit on which the relevant piece of equipment is fitted. This is always a numeric character and is usually 1 for Boiler 1 etc. Where the equipment is common to all areas of the plant (e.g. a cooling tower) the number 0 is allocated to it.

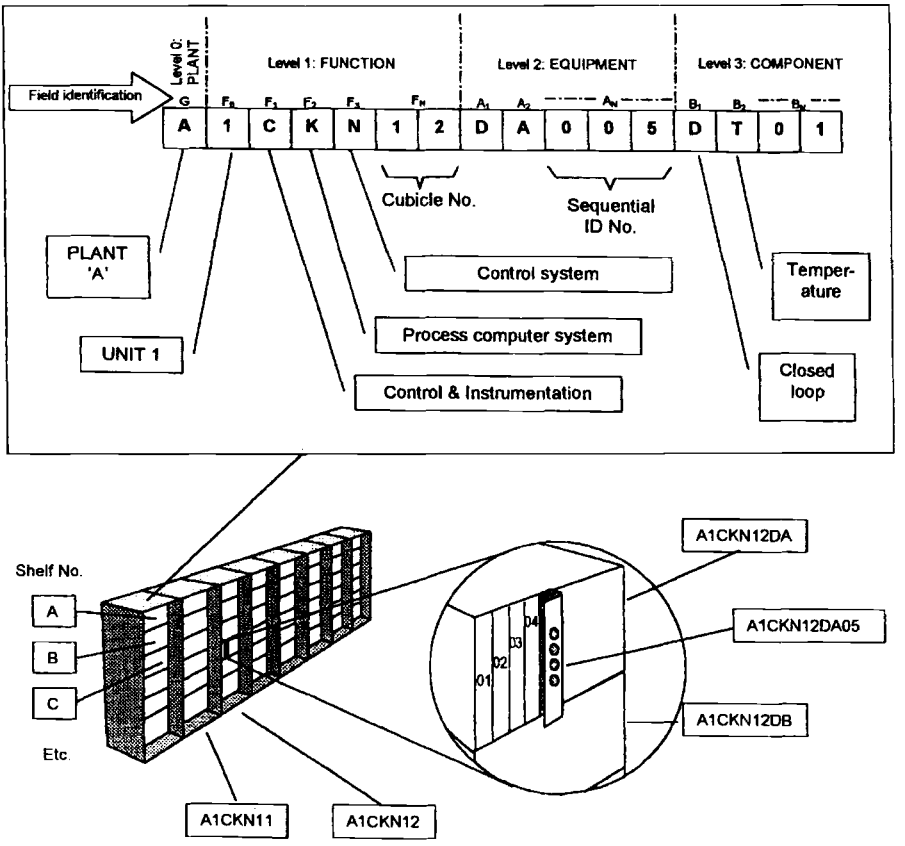
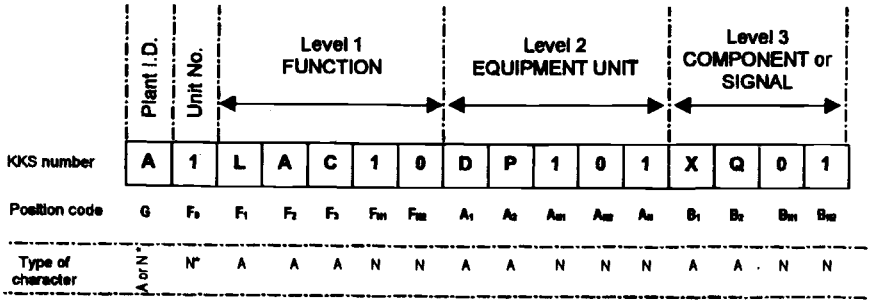



*Figure 9.2 Application of the KKS numbering system to DCS equipment*

- The remaining part of the Level 1 code defines the function of the equipment (e.g. a feed-water pump system, comprising the pump itself, its motor and all equipment associated with it, such as leak-off valve, starter etc.), together with a unique sequential number for the system (e.g. pump system no 2). This group of digits (the first three alphabetic, the next two numeric) is combined with the unit-identifying prefix, into what is called the 'function code'.
- The Level 2 code (which may comprise five or more digits) defines the particular piece of equipment (such as a pressure transmitter) and its unique sequential number (e.g. 101).
- The Level 3 (component or signal) code identifies the component itself, and in the case of a device that generates some form of electrical signal, it also defines the nature of that signal.




A = alphabetic character
N = numeric character
* = optional (may be omitted)

***Breakdown of the code***

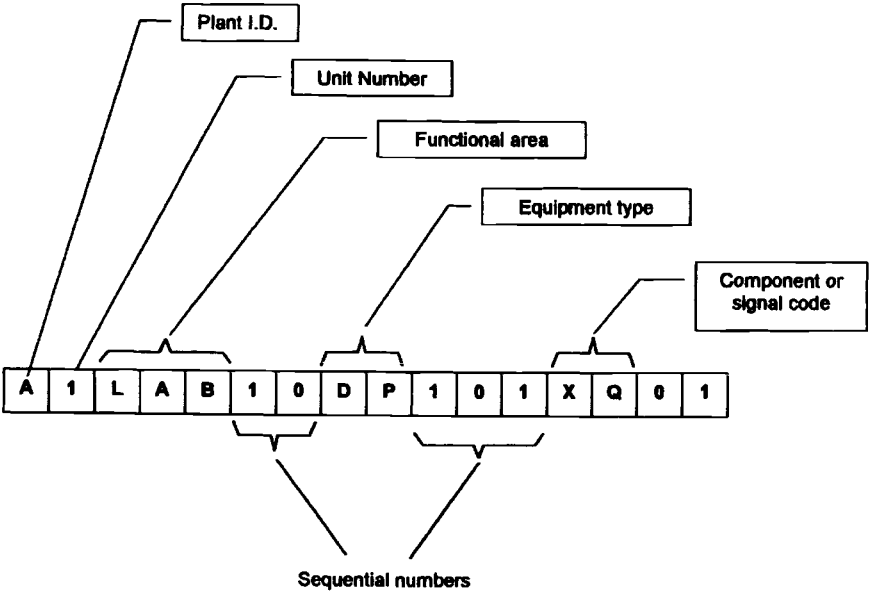



*Figure 9.3 Overall KKS numbering philosophy*

It is important to understand that if there are, for example, 200 pressure gauges on a plant, the sequential numbers for these do not start at 001 and continue through to 200. The numbers relate to the functional area of the plant on which they are used. Therefore, they start afresh each time the preceding code changes. For example the first pressure gauge on the HP steam piping system of a plant (where the piping system is numbered LBA10) will be allocated a sequential number LBA10CP001, the next will be LBA10CP002, and so on. The numbering starts again on another system: the first pressure gauge on the hot reheat steam piping system (numbered LBB10) being allocated the sequential number LBB10CP001, the next being LBB10CP002 and so on.

The structure and application of this coding system is illustrated in Figures 9.2 and 9.3.

### *9.3.2.2 Explanation of the main coding principles*

There are several hundred categories within the KKS codes but the following is a brief summary which will provide some assistance in dealing with boiler C&I systems. Where it is felt that the wording is ambiguous, some examples have been provided. However, it should be remembered that these interpretations are not universally implemented, and the importance of project-specific agreement and co-ordination (as mentioned in Section 9.3.1) cannot be too strongly emphasised.

Level 1: Function

A = grid and distribution systems
B = power transmission and auxiliary power supply
C, D = instrumentation and control equipment
E = conventional fuels supply and residue disposal
F = handling of nuclear equipment
G = water supply and disposal
H = conventional (i.e. nonnuclear) heat generation
J = nuclear heat generation
K = reactor auxiliary systems
L = steam, water and gas cycles
M = main machine sets
N = process energy (e.g. district heating)
P = cooling-water systems
Q = auxiliary systems (e.g. air compressors)
R = gas generation and treatment
S = ancillary systems (e.g. heating and ventilation)
U = structures

W = renewable energy plants
X = heavy machinery
Z = workshop and office equipment

Level 2: Equipment unit
A = mechanical equipment (e.g. valves, dampers, fans, including actuators)
B = mechanical equipment (e.g. storage tanks)
C = direct measuring circuits (e.g. local indicators)
D = closed-loop control circuits
E = analogue and digital signal conditioning
F = indirect measuring circuits (e.g. sensors feeding remote indicators)
G = electrical equipment (e.g. cubicles, junction boxes, generators, inverters, batteries, lightning-protection system)
H = subassemblies of main and heavy machinery (e.g. bearings)
J = nuclear assemblies (e.g. absorbers, moderators, shielding equipment)

### *9.3.3 An example of how the codes are used*

Figure 9.4 shows how the KKS codes may be applied to a part of the feed-water pumping system on a plant. In this example, the pump relates to Boiler 1 and the pressure indicator at its discharge is therefore numbered 1LAB25CP001, the C being the Level 2 code for a direct-measuring device (C) indicating pressure (P). The feed-flow transmitter forms part of the three-element feed-water control system and its Level 2 code therefore uses D for a closed-loop control circuit, and F for flow.

When it comes to transmitters whose signals do not form part of a closed-loop control system, different interpretations are applied by various users to the Level 2 codes. In some cases the classification letter D is applied to any transmitter which feeds the DCS, irrespective of whether or not the signal is used in a closed-loop control system. Other users extend the meaning of F (indirect measuring circuits) to include this type of measurement. In Figure 9.4, the former of these interpretations is used, and the temperature transmitter that is used to provide a signal for the operator display, is therefore allocated the letter D as the first character of its Level 2 code (the second character is T, for temperature).

This point illustrates the importance of co-ordination and agreement between all parties mentioned at the beginning of Section 9.3.1.

When the Level 3 component code is considered, the KKS classification serves two functions. One identifies the component itself, for example QT01 for a thermowell, while the other designates the signal. The signal codes are broken down into classifications which identify whether they are

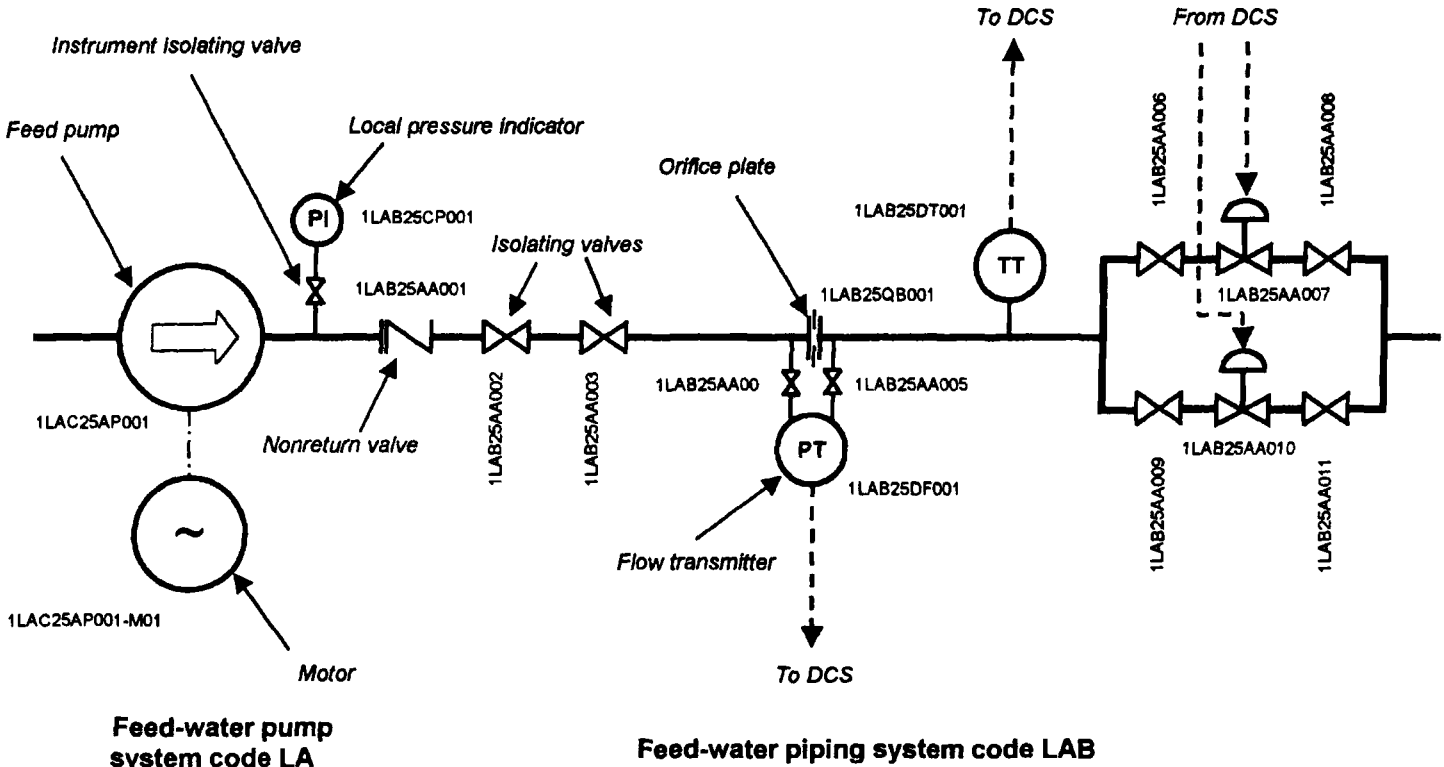



*Figure 9.4 KKS coding applied to a feed-water pump system*

*Table 9.1 Some KKS Level 3 codes for DCS inputs and outputs*

| Function | Field identification $B_1$ | $B_2$ | $B_N$ | Class of signal | Signal |
|---|---|---|---|---|---|
| | | B | 11 | Binary–MCC | Alarm |
| | | | 91 | Binary–MCC | Other |
| | | | 31 | Process switch | LOW limit switch 1 |
| | | | 32 | Process switch | LOW limit switch 2 |
| Input to DCS | X | G | 33 | Process switch | LOW limit switch 3 |
| | | | 41 | Process switch | HIGH limit switch 1 |
| | | | 42 | Process switch | HIGH limit switch 2 |
| | | | 43 | Process switch | HIGH limit switch 3 |
| | | | 91 | Process switch | Other |
| | | | 01 | Analogue input | Resistance thermometer element |
| | | Q | 11 | Analogue input | Transmitter |
| | | | 91 | Analogue input | Other |
| | | B | 01–05 | Binary–MCC | Various START commands |
| | | | 09 | Binary–MCC | STOP |
| | | G | 01 | Binary output | Solenoid valve |
| Output from DCS | Y | | 91 | Binary output | |
| | | | 11 | Analogue output | Speed control |
| | | Q | 21 | Analogue output | Position control |
| | | | 91 | Analogue output | Other |

transmitted to or from the DCS. They can also show the function of the signal. Table 9.1 gives some examples to show how this is achieved.

In general, an X in the $B_1$ field indicates an input to the DCS and Y shows an output from it. In the $B_2$ field, B indicates a binary signal associated with motor controllers; G is for binary signals such as limit switches, pressure switches, solenoid valves and so on; Q is for analogue signals.

By careful use of these identifications it is possible to economise on the work of defining the functions of various signals, since each function will fall into a defined group, for which the actions and connections are already defined.

## 9.4 Summary

This almost completes this study of boiler C&I systems. However, updating and refurbishment needs to be discussed. This is a subject that

has become important as the pace of electronics development has accelerated. As there is little sign of this trend levelling out, we must look at its implications for power-plant management, users and maintenance staff.

## 9.5 References

1 DIN 40719 'Circuit diagrams, circuit representations, definitions, classification (diagrams, charts, tables, item designations)—alphabetical classification examples, 1978